1 Ultrasonics
An Overview

1.1 INTRODUCTION

Viewed from one perspective, one can say that, like life itself, ultrasonics came from the sea. On land, the five senses of living beings (sight, hearing, touch, smell, and taste) play complementary roles. Two of these, sight and hearing, are essential for long-range interaction, while the other three have essentially short-range functionality. But things are different under water; sight loses all meaning as a long-range capability, as does indeed its technological counterpart, radar. So, by default, sound waves carry out this long-range sensing function under water. The most highly developed and intelligent forms of underwater life (e.g., whales and dolphins) over a time scale of millions of years have perfected very sophisticated range finding, target identification, and communication systems using ultrasound. On the technology front, ultrasound also really started with the development of underwater transducers during World War I. Water is a natural medium for the effective transmission of acoustic waves over large distances, and it is indeed, for the case of transmission in opaque media, that ultrasound comes into its own.

In this book, we are more interested in ultrasound as a branch of technology as opposed to its role in nature, but a broad survey of its effects in both areas will be given in this chapter. Human efforts in underwater detection were spurred in 1912 by the sinking of *RMS Titanic* by collision with an iceberg. It was quickly demonstrated that the resolution for iceberg detection was improved at higher frequencies, leading to a push toward the development of ultrasonics as opposed to audible waves. This led to the pioneering work of Langevin, who is generally credited as the father of the field of ultrasonics. The immediate stimulus for his work was the submarine menace during World War I. The United Kingdom and France set up a joint program for submarine detection, and it is in this context that Langevin set up an experimental immersion tank in the Ecole de Physique et Chimie in Paris. He also conducted large-scale experiments, up to 2 km long, in the Seine River. The condenser transducer was soon replaced by a quartz element, resulting in a spectacular improvement in performance, and detection up to a distance of 6 km was obtained. With Langevin's invention of the more efficient sandwich transducer shortly thereafter, the subject was born. Although these developments came too late to be of much use against submarines in that war, numerous technical improvements and commercial applications followed rapidly.

But what, after all, is ultrasonics? Like the visible spectrum, the audio spectrum corresponds to the standard human receptor response function and covers frequencies from 20 Hz to 20 kHz, although, with age, the upper limit is reduced significantly. For both light and sound, the "human band" is only a tiny slice of the total available bandwidth. In each case, the full bandwidth can be described by a complete and unique theory, that of electromagnetic waves for optics and the theory of stress waves in material media for acoustics.

Ultrasonics is defined as that band above 20 kHz. It continues up into the megahertz range and finally, at around 1 GHz, goes over into what is conventionally called the hypersonic regime. The full spectrum is shown in Figure 1.1, where typical ranges for the phenomena of interest are indicated. Most of the applications described in this book take place in the range of 1 to 100 MHz, corresponding to wavelengths in a typical solid of approximately 1 mm to 10 μm, where an average sound velocity is about 5000 m/s. In water—the most widely used liquid—the sound velocity is about 1500 m/s, with wavelengths of the order of 3 mm to 30 μm for the above frequency range.

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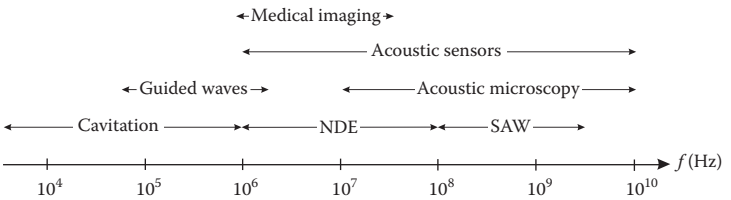

FIGURE 1.1 Common frequency ranges for various ultrasonic processes.

Optics and acoustics have followed parallel paths of development from the beginning. Indeed, most phenomena that are observed in optics also occur in acoustics. But acoustics has something more—the longitudinal mode in bulk media, which leads to density changes during propagation. All the phenomena occurring in the ultrasonic range occur throughout the full acoustic spectrum, and there is no theory that works only for ultrasonics. So the theory of propagation is the same over the whole frequency range, except in the extreme limits where funny things are bound to happen. For example, diffraction and dispersion are universal phenomena; they can occur in the audio, ultrasonic, or hypersonic frequency ranges. It is the same theory at work, and it is only their manifestation and relative importance that change. As in the world of electromagnetic waves, it is the length scale that counts. The change in length scale also means that quite different technologies must be used to generate and detect acoustic waves in the various frequency ranges.

Why is it worth our while to study ultrasonics? Alternatively, why is it worth the trouble to read (or write) a book like this? As reflected in the structure of the book itself, there are really two answers. First, there is still a lot of fundamentally new knowledge to be learned about acoustic waves at ultrasonic frequencies. This may involve getting a better understanding of how ultrasonic waves occur in nature, such as a better understanding of how bats navigate or dolphins communicate. Also, as mentioned later in this chapter, there are other fundamental issues where ultrasonics gives unique information; it has become a recognized and valuable tool for better understanding the properties of solids and liquids. Superconductors and liquid helium, for example, are two systems that have unique responses to the passage of acoustic waves. In the latter case, they even exhibit many special and characteristic modes of acoustic propagation of their own. A better understanding of these effects leads to a better understanding of quantum mechanics and hence to the advancement of human knowledge.

The second reason for studying ultrasonics is because it has many applications. These occur in a very broad range of disciplines, covering chemistry, physics, engineering, biology, food industry, medicine, oceanography, seismology, and so on. Nearly all of these applications are based on two unique features of ultrasonic waves:

1. Ultrasonic waves travel slowly, about 100,000 times slower than electromagnetic waves. This provides a way to display information in time, create variable delay, and so on.
2. Ultrasonic waves can easily penetrate opaque materials, whereas many other types of radiation such as visible light cannot. Since ultrasonic wave sources are inexpensive, sensitive, and reliable, this provides a highly desirable way to probe and image the interior of the opaque objects.

Either or both of these characteristics occur in most ultrasonic applications. We will give one example of each to show how important they are. Surface acoustic waves (SAWs) are high-frequency versions of the surface waves discovered by Lord Rayleigh in seismology. Because of their slow velocity, they can be excited and detected on a convenient length scale (cm). They have become an important part of analog signal processing, for example, in the production of inexpensive, high-quality filters, which now find huge application niches in the television and wireless communication markets. A second example is in medical applications. Fetal images have now become a standard

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part of medical diagnostics and control. The quality of the images is improving every year with advances in technology. There are many other areas in medicine where noninvasive acoustic imaging of the body is invaluable, such as cardiac, urological, and opthalmological imaging. This is one of the fastest growing application areas of ultrasonics.

1.2 ULTRASONICS IN NATURE

We consider first the important case of ultrasonic communication by animals, including mammals, rodents, fish, birds, and so on. In fact, only a limited number of species use this capability and they mostly do so in environments where normal sensing mechanisms are not very effective, for example, night or underwater vision or where there is high background audible noise. The sensing functions include navigation and communication for group interaction or survival such as attracting mates, evading predators, or detecting prey. The relative importance of ultrasonics in the sensing arsenal of the animal will depend on such technical factors as attenuation, scattering, directionality, and so on as compared to audible or optical communication.

There has recently been heightened interest in this field mainly due to the use of highly controlled laboratory experiments. Often this involves surgical intervention to allow direct study of the animal's sensing system, accompanied by Magnetic Resonance Imaging (MRI) mapping of the cervical areas involved in different functions. The laboratory also allows controlled modification of the environment, which gives complementary information to field observations. Development of improved ultrasonic instrumentation, microphones, and so on has also led to more refined experiments. Very recent work has also focused on breakthroughs in the genetic area.

In this short review, we begin with one of the most advanced practitioners of ultrasonics, the bat. This is followed by accounts of evasive measures developed by its prey, such as moths, mantis, and crickets. We then consider rodents, followed by frogs and the cetacean (whales, porpoises, and dolphins).The discussion is rounded out by the most spectacular of all, the snapping shrimp.

One of the best-known examples is ultrasonic navigation by bats, the study of which has a rather curious history [1]. The Italian natural philosopher Lazzaro Spallanzani published results of his work on this subject in 1794. He showed that bats were able to avoid obstacles when flying in the dark, a feat that he attributed to a "sixth sense" possessed by bats. This concept was rejected in favor of a theory related to flying by touch. In the light of further experimental evidence, Spallanzani modified his explanation to one based on hearing. Although this view was ultimately proven to be correct, it was rejected and the touch theory was retained. The subject was abandoned; it was only in the mid twentieth century that serious research was done in the subject, principally by Griffin and Pye. More complete investigations have shown that the bat is fully equipped with an animal sonar system; there are many detailed variations as the technique is used by the suborder *microchiptera*, which contains over 800 species. Emission of the signals is from the larynx at 10–200 pulses per second (pps). A low rate of 10 pps is used far from the target to ensure good resolution and the rate increases to 200 pps close to the target. The duration (0.2–100 ms) and intensity (50–120 dB) are decreased when nearing the target to avoid receiver saturation and temporary deafness. The pulse interval is linked to the desired range (100 ms allows 17 m range, while 5 ms gives a range of 85 cm). Directionality is assured in the usual way by reception with the two ears of the bat. A number of studies have shown that the bat has an incredibly sophisticated signal processing system. There are two main modes of operation: (1) Frequency modulation (FM) sweep, which permits very precise localization by short duration signals. It is effective for close-in work and where there are numerous targets, as it affords a resolution as good as 1 mm. It allows the use of cross-correlation techniques by the bat. The disadvantage is the reduced range due to the use of many frequencies, meaning a lower signal level per frequency; and (2) Constant frequency (CF), which allows the use of Doppler shifting. It is best adapted to a stationary bat in an open environment. Long-range detection is possible due to the concentration of energy at one frequency. Harmonics can also be used and these are also exploited for Doppler. The bat's auditory system is, of course, highly developed to handle

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this sophisticated signal processing. MRI studies have shown that FM and CF modes involve different parts of the bat's brain [2]. In some cases, the physiological system is very specialized for the technique used; the mustached bat has a thickened membrane to optimize reception at 61–61.5 kHz, precisely the Doppler-shifted second harmonic frequency. In summary, there is evidence that the bat's echolocation system is almost perfectly optimized; small bats are able to fly at full speed through wire grid structures that are only slightly larger than their wingspans. Good discussions of echolocation by bats are given in Suga [2] and Fenton [3].

It is also fascinating that one of the bat's main prey, the moth, is also fully equipped ultrasonically. The moth can detect the presence of a bat at great distances—up to 100 ft—by detecting the ultrasonic signal emitted by the bat. Laboratory tests have shown that the moth then carries out a series of evasive maneuvers and sends out an ultrasonic signal consisting of a series of clicks. For a long time, the precise role of these clicks was uncertain, but recent controlled laboratory experiments with tethered tiger moths have shown unambiguously that the clicks represent a jamming signal to be picked up by the bat [4]! Another fascinating example is the praying mantis. It has a single ear with which it detects the approaching bat. Just before the bat arrives, it goes into a steep dive, just like a fighter pilot engaged in a dog fight [5]. Finally, another example of ultrasonics used by insects is provided by the cricket *meconematinae katydid* in the South American rain forest [6]. Most crickets produce sound by flipping their wings, but this katydid produces the highest frequency of all insects by another method. He has a scraper and uses it to close the forewings to store elastic energy. At some point, the scraper slips free and the elastic energy is released in the form of a short ultrasonic pulse. Crickets have another specialty in that the female has an incredibly high directional sensitivity that cannot be explained by the intensity difference received by the two ears. It turns out that the female is sensitive to the phase difference, an effect which is optimized if the male sends out a long pure tone.

Several types of birds use ultrasonics for echolocation, and, of course, acoustic communication between birds is highly developed. Of the major animals, the dog is the only one to use ultrasonics. Dogs are able to detect ultrasonic signals that are inaudible to humans, which is the basis of the silent dog whistle. However, dogs do not need ultrasonics for echolocation, as these functions are fully covered by their excellent sight and sense of smell for long- and short-range detection. Other examples of terrestrial animals using ultrasonics are rodents, chiefly Richardson's ground squirrel, the prairie gopher. These rodents live in colonies and use ultrasonic alarms to warn other members of the presence of predators. They use both audio (8 kHz) and ultrasonic (48 kHz) alarms, depending on the distance of the predator.

Anyone who lives in the country knows that singing is a vital part of a frog's life. Amphibians play a link role between land and aquatic life, and this is also true for animal's use of ultrasonics. The Chinese concave-eared torrent frog, *Odorrana Tormata*, is a very interesting example of a creature's adaptation to its environment. It has been studied in detail by Feng and Narins over the last 10 years [7] and the findings are fascinating. These frogs live near noisy streams, leading to a high ambient noisy background, which was found to cover virtually the full audio spectrum. The frogs get around this by making their calls in the ultrasonic range, producing intense chirps up to 128 kHz. They use all the tricks developed independently by bats: CF and FM calls, period doubling, and sub-harmonics, indicative of nonlinear chaotic behavior. Their recessed eardrums and shortened connecting bones and other physiological aspects allow them to hear up in the ultrasonic range. Most surprisingly, they have active control of the opening to their Eustachian tubes, allowing them to close the tubes when necessary to cut down the background noise to improve the signal-to-noise ratio. Like the katydid crickets, both males and females display an extraordinary localization acuity, even though the distance between their ears is tiny. They also exhibit exceptional navigational ability in darkened laboratory cages. These phenomena are not yet understood in detail but it would seem clear that they are accomplished by the frogs' unusual ultrasonic abilities.

Passing to the use of ultrasonics under water, the seal is another interesting transition story. The seal provides nature's lesson in acoustic impedance, as it has two sets of ears—one set for use in

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air, centered at 12 kHz, and the other for use under water, centered at 160 kHz. These frequencies correspond to those of its principal predators. As will be seen for dolphins and whales, the ultrasonic frequencies involved are considerably higher than those in air; this is necessary to get roughly similar spatial resolution in the two cases, as the speed of sound in water is considerably higher than in air.

Next to bats, dolphins (porpoises) and whales are the best-known practitioners of ultrasound. Their ultrasonic emissions have been studied extensively and the work is ongoing. It is believed that dolphins have a well-defined vocabulary. Some of the sounds emitted are described by graphic terms such as mewing, moaning, rasping, whistling, and clicking, all with characteristic ultrasonic properties. The latter two are the most frequent. The whistle is a low-frequency sound in pulses about a second long and frequencies in the range 7 to 15 kHz. The clicks are at considerably higher frequencies, up to 150 kHz, at repetition rates up to several hundred per second. The widths of the clicks are sufficiently short so that there is no cavitation set up in the water by the high amplitudes that are generated. It is well known that toothed whales (orcas and sperm whales) use echolocation techniques. They have a fatty organ that focuses the emitted sound in front of them. Their receiving ability is improved by layers of fatty tissue in their lower jaw, which are impedance-matched to water; this increases the signal level transmitted to the middle ear. Bio sonar is essential for sperm whales as they feed almost exclusively on squid. The latter's evasive tactics include going to great depths in the sea, where any visual system would be useless for detecting them and ultrasonics becomes essential. These examples illustrate that species generally develop an ultrasonic capability only when it is essential; toothed whales that feed on squid need ultrasonics, while krill-feeding baleen whales do not. Likewise, microbats that hunt insects in the dark need ultrasonics, while fruit bats do not.

We complete the survey of animal acoustics with the incredible story of the snapping shrimp. This shrimp has one normal claw and a huge snapper claw. The latter is held open until prey is lined up, at which point it is closed at lightening speed. This is accompanied by a noise, and it turns out that this is due to cavitation bubbles collapsing, exactly as described in Chapter 12 in the study of single bubble sonoluminescence. The cavitation caused by the snapping shrimp has been studied by the same techniques using a high speed streak camera and the same phenomena, including shock wave formation, have been observed [8].

Very recently, it has been shown on the molecular level that bats and cetaceans developed bio sonar quite independently due to similar sequencing of the gene prestin and their corresponding proteins. Prestin is a hearing gene that is found in the outer hair cells that act as amplifiers for the inner ear. This provides a clear example of independent adaptive convergence and suggests that there are limited paths, or maybe only one, for the development of echolocation in mammals [9].

While the use of ultrasound by animals such as bats and whales is very well known, it is not often realized that natural events can give rise to ultrasonic waves. Earthquakes emit sound, but it is in the very-low-frequency range, below 20 Hz, which is called *infrasound*. The much higher ultrasonic frequencies are emitted in various processes that almost always involve the collapse of bubbles, which is described in detail in Chapter 12. The resonance of bubbles was studied by Minnaert, who calculated the resonance frequency and found that it varied inversely with the bubble size. Hence, very small bubbles have very high resonance frequencies, well into the ultrasonic range. Bubbles and many other examples of physics in nature are described in a charming book, *Light and Color in the Open Air*, by Minnaert [10].

The babbling brook is a good example of ultrasonic emission in nature as the bubbles unceasingly form and collapse. Leighton [11] measured a typical spectrum to be in the range of 3 to 25 kHz. Waterfalls give rise to the high-frequency contact, while low frequencies are produced by the water as it flows over large, round boulders. Another classic example is rain falling on a puddle or lake. The emitted sound can easily be measured by placing a hydrophone in the water. Under usual conditions, a very wide spectrum, 1 to 100 kHz, is obtained, with a peak around 14 kHz. The source of the spectrum is the acoustic emission associated with the impact of water drop on

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the liquid surface and the entrainment of bubbles. It turns out that the broad spectrum is due to the impact and the peak at 14 kHz to the sum of acoustic resonances associated with the bubble formation. An analogous effect occurs with snowflakes that fall on a water surface, apparently giving rise to a deafening cacophony beneath the surface.

Easily the largest source of ultrasound is the surface of an ocean, where breaking waves give rise to a swirly mass of bubbles and agitated water. The situation is, of course, very complicated and uncontrolled, with single bubbles, multibubbles, and fragments thereof continually evolving. This situation has been studied in detail by oceanographers. The effect is always there, but like the tree falling in the forest, there is seldom anyone present to hear it.

1.3 HISTORICAL DEVELOPMENT

Although ultrasonics in nature is a fascinating study in its own right, of far greater interest is the development of the technology of ultrasonic waves that is studied in the laboratory and used in industry. Ultrasonics developed as part of acoustics—an outgrowth of inventions by Langevin. There were, of course, a number of precursors in the nineteenth and early twentieth centuries. In what follows, we summarize the main developments from the beginning until about 1950; this discussion relies heavily on the excellent review article by Graff [1]. After 1950, the subject took off due to a happy coincidence of developments in materials, electronics, industrial growth, basic science, and exploding opportunities. There were also tremendous synergies between technology and fundamental advances. It would be pointless to describe these developments chronologically, and so a sectorial approach is used.

A number of high-frequency sources developed in the nineteenth century were the precursors of things to come. They included the following:

1. The Savant wheel (1830) can be considered the first ultrasonic generator. It worked up to about 24 kHz.
2. The Galton whistle (1876) was developed to test the upper limit of hearing of animals. The basic frequency range was 3 to 30 kHz. Sounds at much higher frequencies were produced, probably due to harmonic generation, as the operation was poorly understood and not well controlled.
3. Koenig (1899) developed tuning forks that functioned up to 90 kHz. Again, these experiments were poorly understood and the conclusions were erroneous, almost certainly due to nonlinear effects.
4. Various high-power sirens were developed, initially by Cagniard de la Tour in 1819. These operated below ultrasonic frequencies but had an important influence on later ultrasonic developments.

In parallel with the technological developments mentioned earlier, there was an increased understanding of acoustic wave propagation, including velocity of sound in air (Paris 1738), iron (Biot 1808), and water (Calladon and Sturm 1826)—the latter a classic experiment carried out in Lake Geneva. The results were reasonably consistent with today's known values—perhaps understandably so, as the measurement is not challenging because of the low value of the velocity of sound compared with the historical difficulties of measuring the velocity of light. Other notable advances were the standing wave approach for gases (Kundt 1866) and the stroboscopic effect (Toepler 1867), which led to Schlieren imaging.

One of the key events leading directly to the emergence of ultrasonics was the discovery of piezoelectricity by the Curie brothers in 1880; in short order, they established both the direct and the inverse effect, that is, the conversion of an electrical signal to a mechanical signal and vice versa. The twentieth century opened with the greatest of all acousticians, Lord Rayleigh (John W. Strutt). Rayleigh published what was essentially the principia of acoustics, *The Theory of Sound*, in

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1889 [12]. He made definitive studies and discoveries in acoustics, including atomization, acoustic surface (Rayleigh) waves, molecular relaxation, acoustic pressure, nonlinear effects, and bubble collapse.

The sinking of the *Titanic* and the threat of German submarine attacks led to Langevin's experiments in Paris in 1915—the real birth of ultrasonics. On the one hand, his work demonstrated the practicality of pulse echo work at high frequencies (150 kHz) for object detection. The signals were so huge that fish placed in the ultrasonic immersion tank were killed immediately when they entered the ultrasonic beam. On the other hand, the introduction of quartz transducers and then the sandwich transducer (steel–quartz–steel) led to the first practical and efficient use of piezoelectric transducers. Quite surprisingly, almost none of Langevin's work on ultrasonics was published. His work was followed up by Cady, which led to the development of crystal-controlled oscillators based on quartz.

Between the wars, the main thrust was in the development of high-power sources, principally by Wood and Loomis. For example, a very-high-power oscillator tube in the range 200 to 500 kHz was developed and applied to a large number of high-power applications, including radiation pressure, etching, drilling, heating, emulsions, atomization, chemical and biological effects, sonoluminescence, sonochemistry, and so on. Supersonic was the key buzzword, and high-power ultrasonics was applied to a plethora of industrial processes. However, this was mainly a period of research and development, and it was only in the period following this that definitive industrial machines were produced. This period, 1940–1955, was characterized by diverse applications, some of which include the following:

1. New materials, including poled ceramics for transduction
2. The Mason horn transducer (1950) for efficient concentration of ultrasonic energy by the tapered element
3. Developments in bubble dynamics by Blake, Esche, Noltink, Neppiras, Flynn, and others
4. Ultrasonic machining and drilling
5. Ultrasonic cleaning: GE produced a commercial unit in 1950
6. Ultrasonic soldering and welding, advances made mainly in Germany
7. Emulsification: dispersal of pigments in paint, cosmetic products, dyes, shoe polish, and so on
8. Metallurgical processes, including degassing melts

From the 1950s onward, there were so many developments in so many sectors that it is feasible to summarize only the main developments by sector. Of course, the list is far from complete, but the aim is to give examples of the explosive growth of the subject rather than provide an encyclopedic coverage of the developments. The proceedings of annual or biannual conferences on the subject, such as the IEEE Ultrasonics Symposium and Ultrasonics International, are good sources of progress in many of the principal directions.

1.4 PHYSICAL ACOUSTICS

A key element in the explosive growth of ultrasonics for electronic device applications and material characterization in the 1960s and beyond was the acceptance of ultrasonics as a serious research and development (R&D) tool by the condensed matter research community. Before 1950, ultrasonics would not have been found in the toolkit of mainline condensed matter researchers, who relied mainly on conductivity, Hall effect, susceptibility, specific heat, and other traditional measurements used to characterize solids. However, with developments in transducer technology, electronic instrumentation, and the availability of high-quality crystals, it then became possible to carry out quantitative experiments on velocity and attenuation as a function of magnetic field, temperature, frequency, and so on, and to compare the results with the predictions of microscopic theory. The trend continued and strengthened, and ultrasonics soon became a choice technique for condensed

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matter theorists and experimentalists. A huge number of sophisticated studies on semiconductors, metals, superconductors, insulators, magnetic crystals, glasses, polymers, quantum liquids, phase transitions, and many others were carried out, and unique information was provided by ultrasonics. Some of this work has become classic. Two examples will be given to illustrate the power of ultrasonics as a research tool.

Solid state and low-temperature physics underwent a vigorous growth phase in the 1950s. One of the most spectacular results was the resolution of the 50-year-old mystery of superconductivity by the Bardeen, Cooper, and Schrieffer (BCS) theory in 1957. The BCS theory proposed that the conduction electrons participating in superconductivity were coupled together in pairs with equal and opposite momentum by the electron–phonon interaction. The interaction with external fields involves so-called coherence factors that have opposite signs for electromagnetic and acoustic fields. The theory predicted that at the transition temperature there would be a peak of the nuclear spin relaxation time and a straight exponential decrease of the ultrasonic attenuation with temperature. This was confirmed by experiment and was an important step in the widespread acceptance of the BCS theory. The theory of the ultrasonic attenuation was buttressed on the work of Pippard, who provided a complete description of the interaction of ultrasonic waves with conduction electrons around the Fermi surface of metals.

A second example is provided by liquid helium, which undergoes a transition to the superfluid state at 2.17 K. Ultrasonic experiments demonstrated a change in velocity and attenuation below the transition. Perhaps more importantly, further investigation showed the existence of other ways of propagating sound in the superfluid state in different geometries—so that one talks of a first (ordinary), second, third, and fourth sound in such systems. These acoustics measurements went a long way to providing a fuller understanding of the superfluid state. The case of He^3 was even more fruitful for acoustic studies. The phase diagram was much more complicated, involving the magnetic field, and many new hydrodynamic quantum modes were discovered. Recently, even purely propagating transverse waves were found in this superfluid medium.

This and other fundamental work led to attempts to increase the ultrasonic frequency. Coherent generation by application of microwave fields at the surface of piezoelectrics raised the effective frequency well into the hypersonic region above 100 GHz. Subsequently, the superconducting energy gap of thin films was used to generate and detect high-frequency phonons at the gap frequency, extending the range to the terahertz region. Heat pulses were used to generate very-high-frequency broadband pulses of acoustic energy. In another approach, the development of high-flux nuclear reactors led to measurement of phonon dispersion curves over the full high-frequency range, and ultrasonics became a very useful tool for confirming the low-frequency slope of these curves. In summary, all of this work in physical acoustics gave new legitimacy to ultrasonics as a research tool and stimulated development of ultrasonic technologies.

1.5 LOW-FREQUENCY BULK ACOUSTIC WAVE APPLICATIONS

This main focus of our discussion on the applications of ultrasonics provides some of the best examples of ultrasonic propagation. The piezoelectric transducer itself led to some of the earliest and most important applications. The quartz resonator was used in electronic devices starting in the 1930s. The quartz microbalance became a widely used sensor for the detection of mass loading of molecular species in gaseous and aqueous media and will be fully described in Chapter 16. Many other related sensors based on this principle were developed and applied to many problems such as flow sensing (including Doppler), level sensing, and propagation (rangefinders, distance, garage door openers, camera rangefinders, etc.). A new interest in propagation led to the development of ultrasonic nondestructive evaluation (NDE). Pulse echo techniques developed during World War II for sonar and radar led to NDE of materials and delay lines using the same principles and electronic instrumentation. NDE of materials with shorter pulse and higher frequencies was made possible with the new electronics developed during the war, particularly radar. A first ultrasonic flaw

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detection patent was issued in 1940. From 1960 to the present, there have been significant advances in NDE technology for detecting defects in multilayered, anisotropic samples, raising ultrasonics to the status of a major research tool, complementary to resistivity, magnetization, x-rays, eddy currents, and so on.

One of the most important areas in low-frequency bulk acoustic wave (BAW) work was the development of ultrasonic imaging, which started with the work of Sokolov. By varying the position and angle of the transducer, A (line scan), B (vertical cross-section), and C (horizontal cross-section) scans were developed. C scan has turned out to be the most commonly used, where the transducer is translated in the *x-y* plane over the surface of a sample to be inspected so that surface and subsurface imaging of defects can be carried out. Realization by Quate in the early 1970s that microwave ultrasonic waves in water have optical wavelengths led to the development of the scanning acoustic microscope (SAM) by Lemons and Quate in 1974. This is covered in detail in Chapter 17 because it is a textbook example of the design of an ultrasonic instrument. The SAM provides optical resolution for frequencies in the gigahertz range, high intrinsic contrast, quantitative measure of surface sound velocities, and subsurface imaging capability. In more recent developments, the atomic force microscope, also developed by Quate, has been used to carry out surface, near-surface, and near-field imaging with nanometer resolution. In parallel, much progress has been made in acoustic imaging with phased arrays. Recent developments include time-reversal arrays and the use of high-performance micromachined capacitive transducer arrays.

1.6 SURFACE ACOUSTIC WAVES

The SAW was one of the modes discovered very early on by Lord Rayleigh in connection with seismology studies. In the device field, it remained a scientific curiosity with few applications until the development of the interdigital transducer by White and Voltmer in the 1960s. This breakthrough allowed the use of planar microelectronic technology, photolithography, clean rooms, and so on for the fabrication of SAW devices in large quantities. A second breakthrough was a slow but ultimately successful development of sputtering of high-quality ZnO films on silicon, which liberated device design from bulk piezoelectric substrates and permitted integration of ultrasonics with silicon electronics. Since the 1960s, there has been a huge amount of work on the fundamentals and the technology of SAW and its application to signal processing, NDE, and sensors. The SAW filter has been particularly important commercially in mass consumer items such as television filters and wireless communications. There is presently a push to very-high-frequency devices (5 to 10 GHz) for communication applications.

The above topics are the main ones covered in the applications sections. Of course, there are many other extremely important areas of ultrasonics, but a selection was made of those topics that seemed best suited as examples of the basic theory and which the author was qualified to address. Some of the important areas omitted (and the reasons for omission) include piezoelectric materials, transducers, medical applications (specialized and technical), high-power ultrasonics (lacks a well-developed theoretical base), underwater acoustics, and seismology (more acoustics than ultrasonics and lacking unity with the other topics). In these cases, a brief summary of some of the highlights is given to complete the introductory survey of this chapter.

1.7 PIEZOELECTRIC MATERIALS

Much of the remarkable progress made in ultrasonics is due to the synergy provided by new high-performance materials and improved electronics. This is perhaps best exemplified in the work of Langevin in applying quartz to transduction and then developing the composite transducer. A second major step forward occurred in the 1940s with the development of poled ceramic transducers of the lead zirconate (PZT) family, which were relatively inexpensive, rugged, with high performance, and ideally suited to field work. For the laboratory, more expensive but very high-performance new

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crystals such as lithium niobate entered into widespread use. A third wave occurred with piezoelectric films. After a false start with CdS, ZnO and AlN became the standard piezoelectric films for device applications such as SAW. The development of polyvinylidine and then copolymers based on it was important for many niche applications—particularly in medical ultrasonics, as the acoustic impedance is very well matched to that in water. Other favorable properties include flexibility and wide bandwidth. They are, however, very highly attenuating, so they are not suitable for SAW or high-frequency applications.

More recently, the original PZT family has been improved by the use of finely engineered piezocomposites for general BAW applications. New SAW substrates are still under development, particularly with the push to higher frequencies. Microelectromechanical transducers are under a stage of intense development as they have potential for high-quality, real-time, mass-produced acoustic imaging systems.

1.8 HIGH-POWER ULTRASONICS

This was one of the first areas of ultrasonics to be developed, but it has remained poorly developed theoretically. It involves many heavy-duty industrial applications, and often the approach is semi-empirical. Much of the early work was carried out by Wood and Loomis, who developed a high-frequency, high-power system and then used it for many applications. One of the problems in the early work was the efficient coupling of acoustic energy into the medium, which limited the available power levels. A solution was found with the exponential horn; a crude model was developed by Wood and Loomis, and this was perfected by Mason using an exponential taper in 1950. The prestressed ceramic sandwich transducers also were important in raising the acoustic power level. Another problem, which led in part to the same limitation, was cavitation. Once cavitation occurs at the transducer or horn surface, the transfer of acoustic energy is drastically reduced due to the acoustic impedance mismatch introduced by the air. However, work on cavitation gradually led to it becoming an important subject in its own right. Ramification of the process led to operations such as drilling, cutting, and ultrasonic cleaners. Other applications of cavitation included sonochemistry and sonoluminescence. High-power ultrasonics also turned out to be a useful way to supply large amounts of heat, leading to ultrasonic soldering and welding of metals and plastics.

1.9 MEDICAL ULTRASONICS

From a purely technical ultrasonic standpoint, there are many similarities between NDE and medical ultrasonics. Basically, one is attempting to locate defects in an opaque object; the same technological approaches are relevant, such as discriminating between closely spaced echoes and digging signals out of the noise. So it is not surprising that many developments on one side have been applied to problems on the other. Of course, there are differences: one is that inspection of *in vivo* samples is an important part of medical ultrasonics. Respiratory effects, blood flow, and possible tissue damage are issues that are totally absent in NDE. This has led to much R&D on induced cavitation and cavitation damage as well as development of very sophisticated Doppler schemes for monitoring blood flow.

Historically, during the 1940s and 1950s, there was strong emphasis on therapy. This declined in the 1950s when the current dominant theme of medical imaging started. There was much work on the brain, followed by applications in urology, ophthalmology, and vital organs (heart and liver). Certainly, the most celebrated application of ultrasonic imaging in medicine is fetal imaging; images of tremendous detail and clarity can be obtained in real time. High-resolution in vitro imaging has been carried out in the same way. Current trends for *in vivo* imaging include phased arrays for real-time imaging and nonlinear imaging using contrast agents as well as harmonic imaging of basic tissue.

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1.10 ACOUSTO-OPTICS

The interaction of light and sound was discovered early in the history of ultrasonics. Brillouin suggested the existence of Brillouin scattering in 1922, which was followed by low-frequency diffraction (Debye-Sears 1932 and Raman-Nath 1935). Schlieren visualization of ultrasonic fields has long been a useful tool for exploring scattering and propagation phenomena. Bragg cells for acousto-optic modulators are important components in optical communication systems. An important developing area is that of laser ultrasonics. It has been known since the 1960s that absorption of a laser beam can lead to generation of ultrasonic waves by the thermoelastic effect. The mode generated can be partly controlled by the surface condition. An all-optical system can be made by using a Michelson interferometer to monitor surface displacement. Applications of laser ultrasonics are described in Chapter 20.

1.11 UNDERWATER ACOUSTICS AND SEISMOLOGY

Fascinating as they are, underwater acoustics and seismology cannot be properly put under the umbrella of ultrasonics as almost all the work in these areas is done in the audio or infrasonic frequency range. It is only the tail end, as it were, of a few graphs that penetrate into the ultrasonic regime. Nevertheless, the basic theory is the same, and only the length scale is much larger. Also, the acoustic phenomena of interest are in many cases identical. One needs only cite the names of Rayleigh, Love, and Sezawa waves in the earth's crust, longitudinal and transverse wave propagation in the bulk of the earth, and multilayer reflection and transmission phenomena in the case of seismology. For underwater acoustics, we have again reflection and transmission phenomena, guided waves in channels due to stratified layers caused by temperature gradients, scattering of acoustic waves by targets of all sorts, bubble phenomena, acoustic imaging, sonar, and the list goes on. In both cases, we have the inverse problem that is at the base of a large chunk of NDE. One of the advantages of the situation, at least in principle, is that it should be relatively easy for experts in ultrasonics to work on problems in these other fields and vice versa.

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